

WEEK 3 STATUS REPORT: LAYER 2 SECURITY & AUTOMATION PIPELINE BASELINE

1. Infrastructure Security Hardening

- 1 Implemented Layer 2 MAC-limiting parameters via Switch Port-Security. Maximum secure address ceiling set to 1 MAC per port with immediate shutdown triggers upon violation.
- 2 Deactivated all unassigned switch ports (Fa0/5 - Fa0/24) via hard administrative shutdown states to mitigate rogue attachment vulnerabilities.
- 3 Initialized Standard Layer 3 Access Control Lists (ACL 1) on all core routers, explicitly binding the VTY management lines to reject SSH handshake requests not originating from the secure server node (192.168.1.10).

2. Automation Architecture Baseline

- 1 Successfully drafted the core initialization structure of 'backup_script.py'.
- 2 Integrated the Python Paramiko SSH Client library framework to orchestrate secure socket contexts against targets R1, R2, and R3.
- 3 Established automated calendar tracking variables to handle timestamp configurations dynamically.